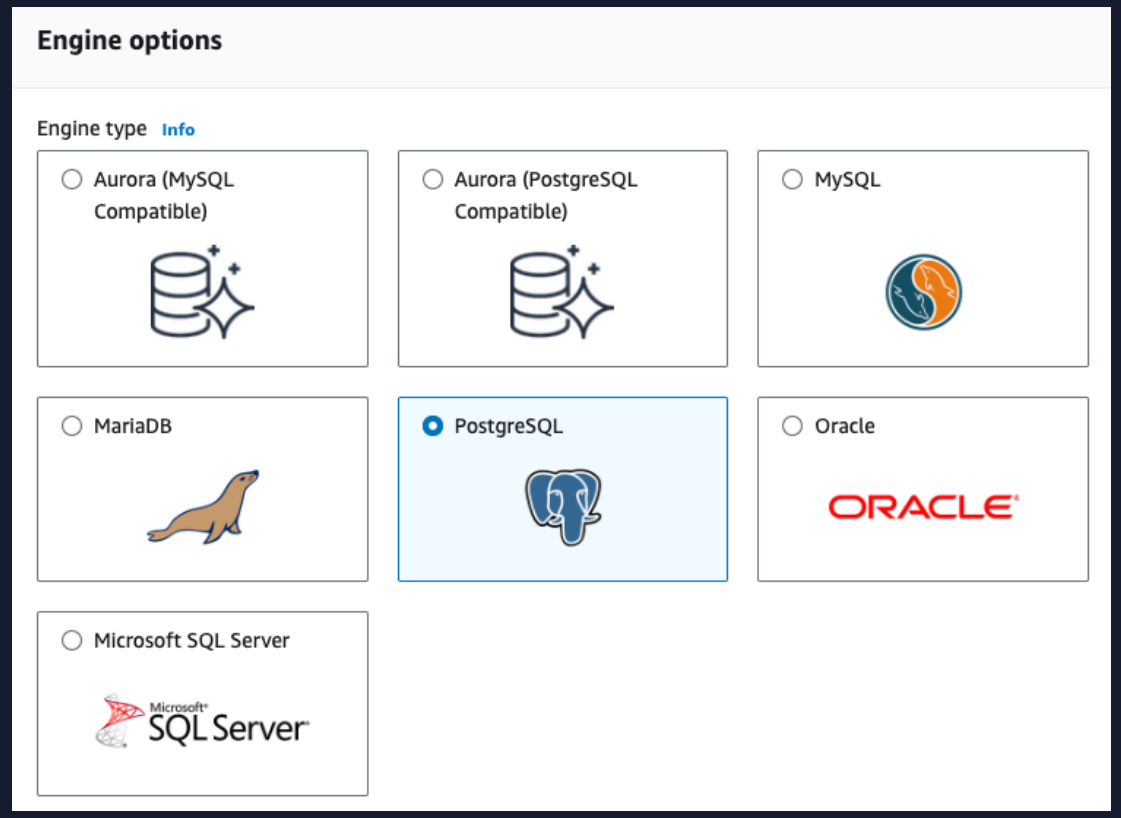
Amazon Web Services: Databases

Topics

- Relational Databases
- OLTP and OLAP
- RDS Advantages
- Multi-AZ
- Unplanned Failures and Maintenance
- Increasing Read Performance
- Amazon Aurora
- Amazon DynamoDB
- Amazon Neptune

Relational Databases

- Have been around for many years and are the heart of many applications
- Consist of tables, rows, and columns
- AWS offers the following in terms of RDS ->
- When would you use an RDS database?
  - online transaction processing (OLTP) workloads



OLTP and OLAP

Understanding the difference between OLTP and OLAP

OLTP

- Processes data from transactions in real time
- E.g., customer orders, payments, booking systems, and so on
- RDS

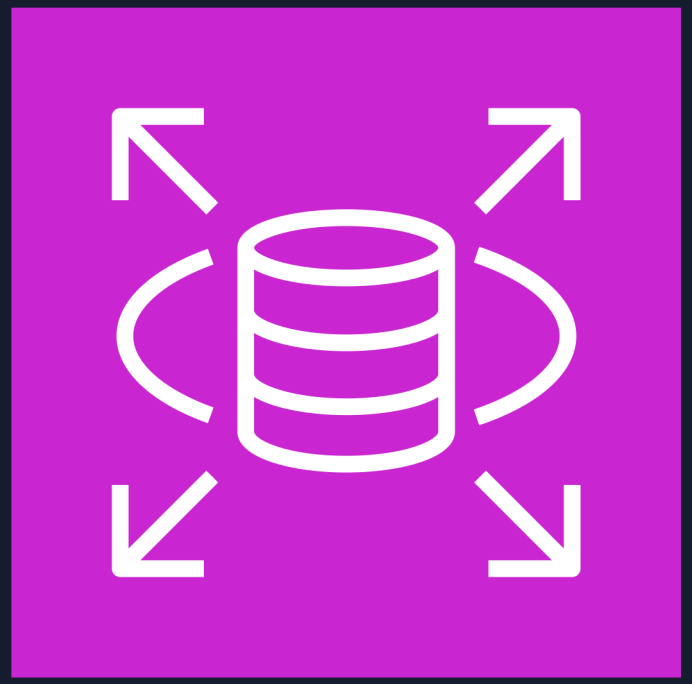
VS

OLAP

- Process complex queries to analyze historical data
- E.g., sales forecasting, analyzing net profit from past 3 years
- Data warehouse such as Redshift

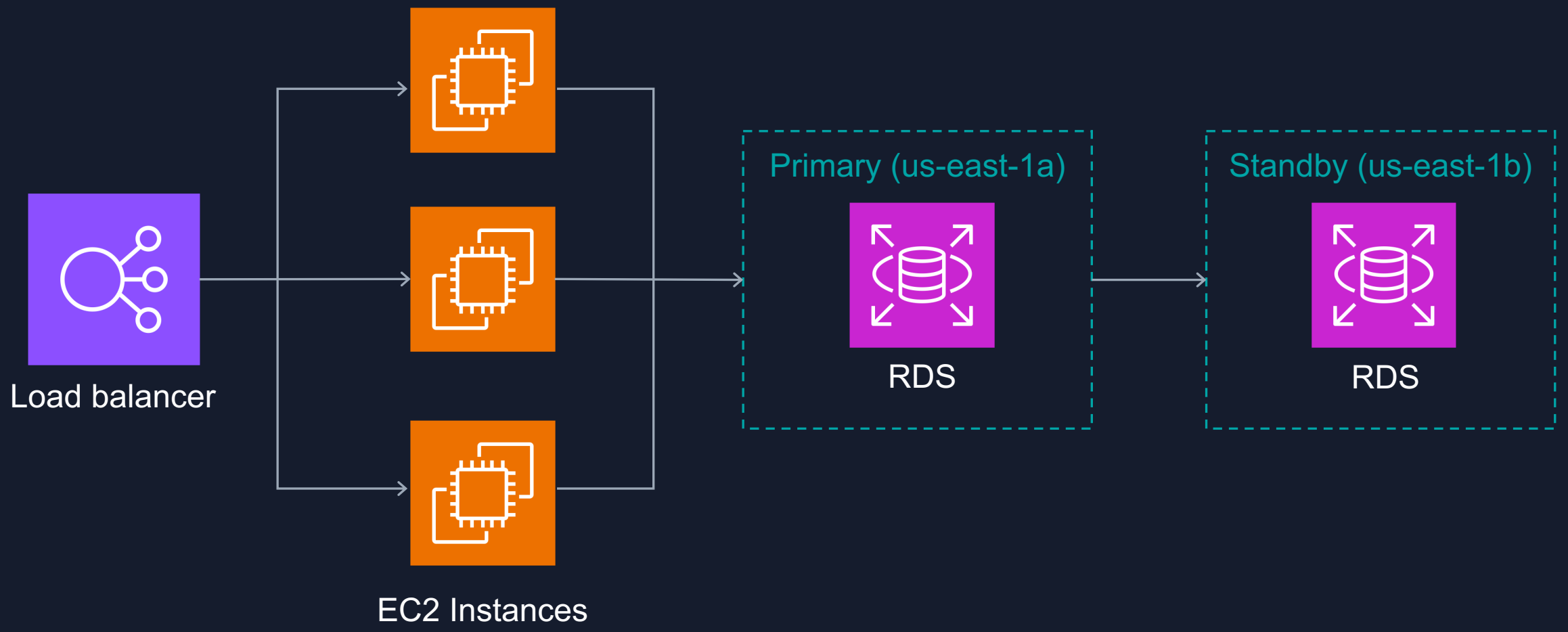
RDS Advantages

- Up and running in minutes
- Multi-AZ
- Failover Capability
- Automated backups



Amazon Relational Database Service (Amazon RDS)

RDS Multi-AZ


Load balancer
EC2 Instances
Primary (us-east-1a)
RDS
Standby (us-east-1b)
RDS

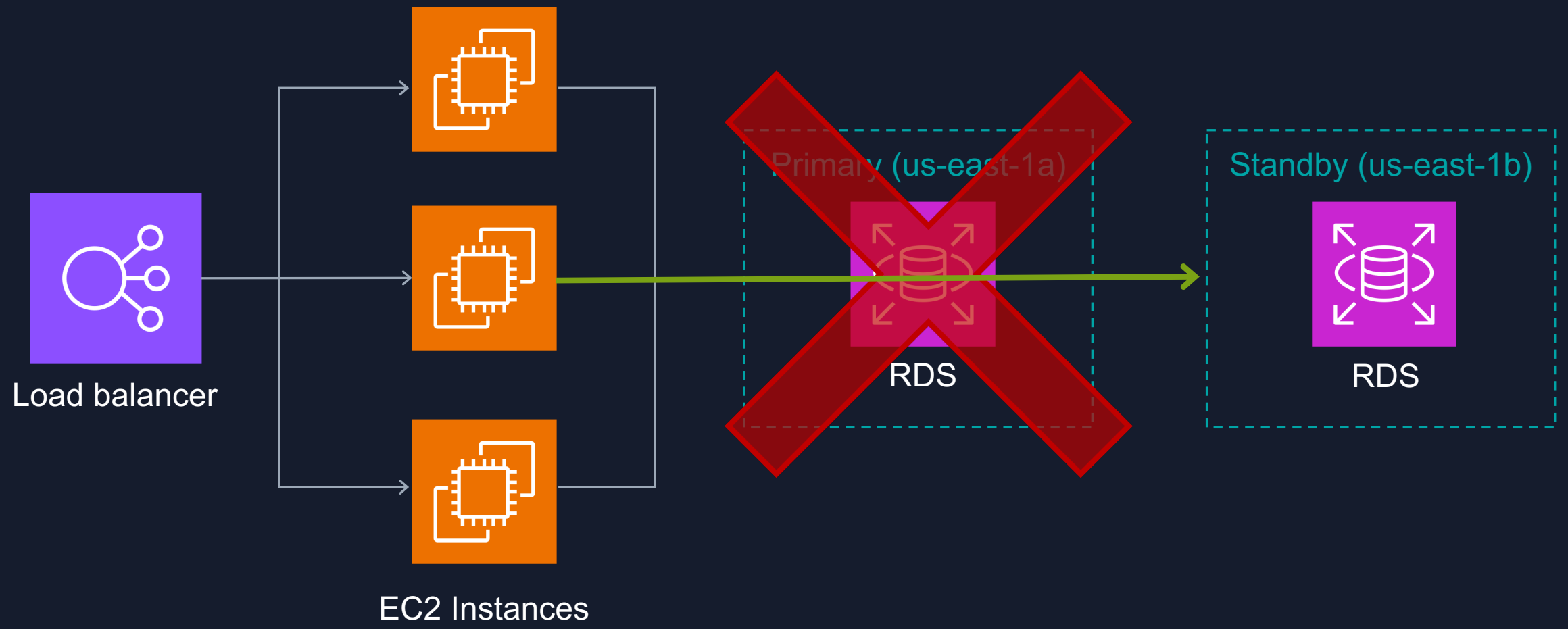
RDS Multi-AZ

- AWS handles replication for you
  - Write to a production DB will automatically synchronize to the standby DB
- RDS types that can be configured as Multi-AZ:

- Aurora is always multi-AZ

Unplanned Failure or Maintenance

- Multi-AZ is for disaster recovery, not for improving performance
- Cannot connect to the standby DB when primary is active



Demo: Improving Performance Using Read Replicas

- A read replica is a read-only copy of your primary DB
- Used for read-heavy workloads to scale read performance
- Takes load off the primary DB
- Can be cross-AZ or cross-Region
- Each read-replica has its own DNS endpoint
- Can be promoted to a primary DB which breaks the replication
- Requires automatic backup to be enabled to deploy a read replica
- Multiple read replicas are supported (up to 5 read replicas to each DB instance)

Amazon Aurora

- Amazon’s proprietary DB
- MySQL and PostgreSQL compatible relational DB engines
- Provides up to 5x better performance than MySQL and 3x better than PostgreSQL DBs
- Provides storage auto-scaling
  - Starts with 10 GBs and scales in 10 GB increments up to 128 Terabytes
- Compute resources can scale up to 96 vCPUs and 768 GB of memory
- You always have 6 copies of your data
  - Can handles the loss of up to 2 copies of data without affecting write availability and up to 3 copies without affecting read availability
- Self-healing storage

Types of Aurora Replicas

Aurora Replicas

Can have 15 read replicas with Aurora

MySQL Replicas

Can have 5 read replicas with Aurora MySQL

PostgreSQL

Can have 5 read replicas with Aurora PostgreSQL

Backup with Aurora

- Automated backup are always enables
- Can take snapshots with Aurora
- Can share snapshots with other AWS Accounts

Aurora Serverless

- On-demand, auto-scaling configuration for the MySQL compatible and PostgreSQL compatible editions of Amazon Aurora
- Serverless DB cluster that scales automatically
- Use case:
  - infrequent,
  - intermittent, or
  - unpredictable workloads

Amazon DynamoDB

- NoSQL DB technology for applications that need consistent, single-digit millisecond latency at any scale
- Supports document and key-value data models
- Great for mobile, web, gaming, ad-tech, IoT, and other types of applications
- Four main facts:
  - Stored on SSD storage
  - Spread across 3 geographically distinct data centers
  - Eventually consistent reads (default)
  - Strongly consistent reads

Difference Between Eventually Consistent Reads and Strongly Consistent Reads

Eventually

- Consistency across all copies of data is usually reached within a second
- Best read performance

VS

Strongly

- Returns a result that reflects all writes that received a successful response prior to the read

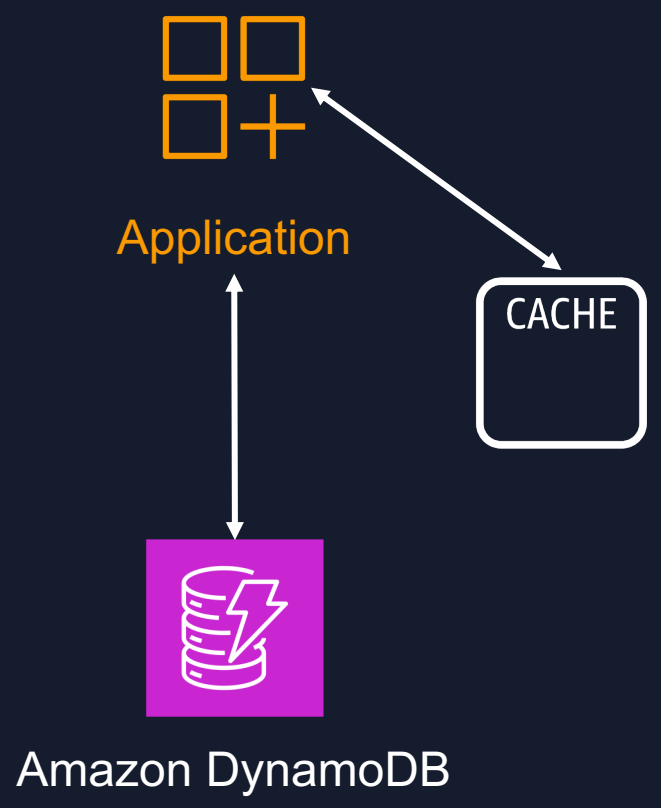
DynamoDB Accelerator (DAX)

- Fully managed, highly available, in memory-cache
- 10x performance improvement
- Reduces request time from milliseconds to microseconds – even under load
- Developers don’t need to manage caching logic – built-in
- Compatible with DynamoDB APIs

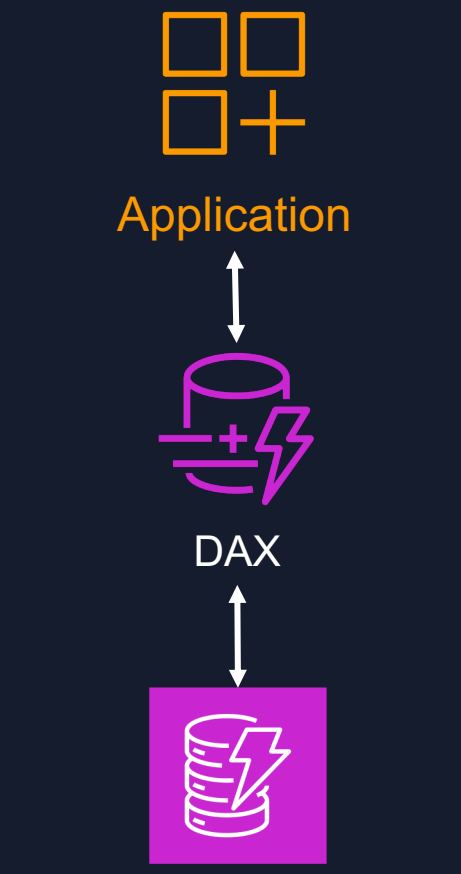
Amazon DynamoDB Accelerator (DAX)

DynamoDB Accelerator (DAX)

Traditional Cache



Caching with DAX



Amazon DynamoDB

On-Demand Capacity

- Pay-per-request pricing
- Balance cost and performance
- No minimum capacity
- Pay more per request than with provisioned capacity
- Use for new product launches

Security

- Encryption at rest using KMS
- Site-to-site VPN
- Direct Connect
- IAM policies and roles
- Fine-grained access
- CloudWatch and CloudTrail
- VPC endpoints

DynamoDB Transactions

ACID for Databases

- Atomic
  - All changes to the data must be performed successfully or not at all
- Consistent
  - Data must be in a consistent state before and after the transactions
- Isolated
  - No other process can change the data while the transaction is running
- Durable
  - The changes made by a transaction must persist

ACID with DynamoDB

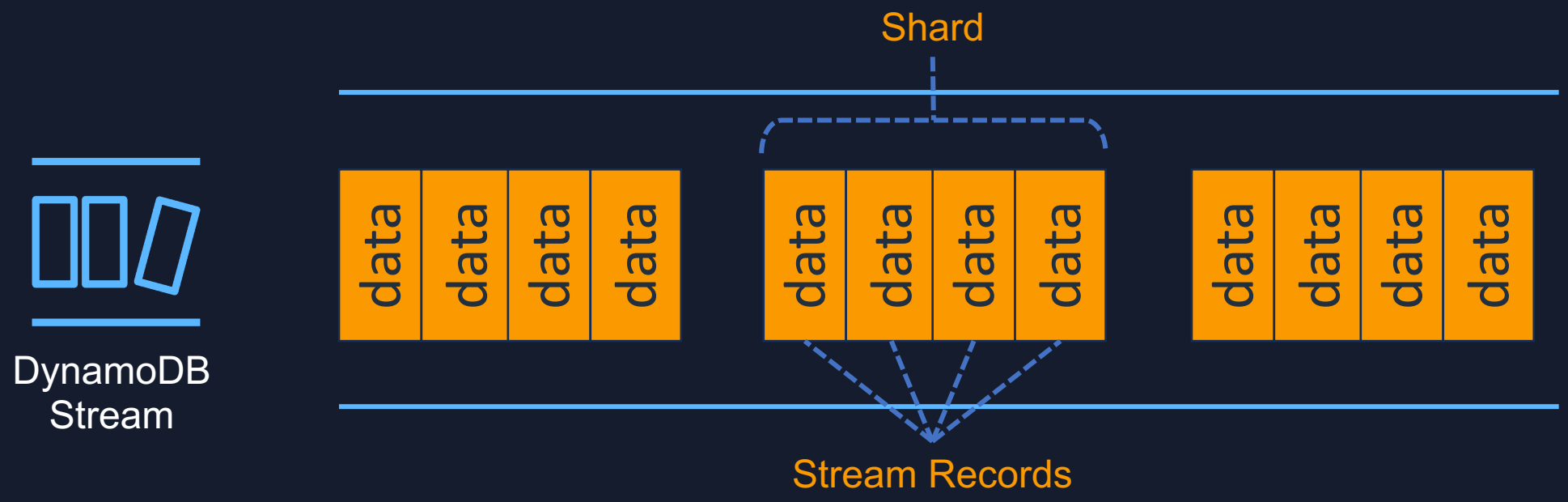
- DynamoDB transactions provide developers ACID across 1 or more tables within a single AWS account and region
- Use transactions when building applications that require coordinated inserts, deleted, or updates as part of a single logical business transaction
- Use cases: financial transactions, fulfilling and managing orders, building multiplayer game engines, and so on

DynamoDB On-Demand Backup and Restore

- Full backups at any time
- Zero impact on table performance and availability
- Consistent within seconds and retained until deleted
- Operates within the same region as the source table
- Point-in-Time Recovery (PITR)
  - Protects against accidental writes or deletes
  - Restore to any point in the last 35 days
  - Incremental backups
  - Not enabled by default

DynamoDB Streams and Global Tables

- Time-ordered sequence of item-level changes in a table
- Stored for 24 hours



DynamoDB Streams and Global Tables

- Managed multi-master, multi-region replication
- Globally distributed applications
- Based on DynamoDB streams
- Multi-region redundancy for disaster recovery or high availability
- No application rewrites
- Replication latency under 1 second

DynamoDB Demo

Amazon Neptune

- Graph database: graph database stores node and relationships instead of tables or documents
- Fast, reliable, fully managed graph DB service
- Use cases:
  - build connection between identities
  - build knowledge graph applications
  - detect fraud patterns
  - Security graphs to improve IT security posture

Checkpoint

What would you use Amazon Neptune for?

1. As a monitoring service
2. As a DNS service
3. As a graph database service
4. To manage cloud infrastructure

What should you do to scale out an RDS database that has a read-heavy workload?

1. Scale RDS vertically.
2. Add additional read replicas.
3. Enable Multi-AZ.
4. Set up RDS automatic backups.

Which of the following is NOT a supported RDS database engine?

1. MariaDB
2. Oracle
3. SQL Server
4. IBM Db2

True or False? RDS read replicas have their own endpoints.

1. false
2. true

Amazon Aurora is compatible with which of the following database engines?

Choose 2

1. MySQL
2. Oracle
3. MariaDB
4. Postgres

DynamoDB is a ____ database.

1. Non-Relational
2. Relational
3. Graph
4. Relationship

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